

GNN-Based Network Intrusion Detection

Evaluation Results — XGBoost vs. E-GraphSAGE vs. HeteroGNN vs. EGAT — UNSW-NB15 and CICIDS Test Sets

Summary — All Models Across Datasets

Dataset	Model	Best Val F1	Test Macro-F1	Weighted-F1	MCC	Kappa	Train Time	Status
UNSW	XGBoost	0.5647	0.5866	0.9836	0.8538	0.8537	18.0 min	Complete
	E-GraphSAGE	0.5097	0.5022	0.9815	0.8279	0.8276	9.2 min	Complete
	HeteroGNN	0.5013	0.5074	0.9811	0.8286	0.8284	9.1 min	Complete
	EGAT	0.3997	0.3734	0.9669	0.7153	0.7133	7.6 min	Complete
CICIDS	XGBoost	0.7000	0.6331	0.9863	0.9467	0.9463	129.2 min	Complete
	E-GraphSAGE	0.6965	0.6380	0.9835	0.9309	0.9305	79.5 min	Complete
	HeteroGNN	0.7041	0.6244	0.9804	0.9057	0.9053	76.8 min	Complete
	EGAT	—	—	—	—	—	—	Failed (CUDA OOM)

Accuracy is reported per model inside each dataset section but is omitted here since it is misleading under severe class imbalance; Macro-F1 and per-class F1 are the primary comparison metrics. Four modeling approaches were evaluated: a gradient-boosted tree baseline (XGBoost) and three GNN architectures (E-GraphSAGE, HeteroGNN, EGAT). EGAT's CICIDS run did not complete due to a CUDA out-of-memory error (see CICIDS section).

UNSW — UNSW-NB15

XGBoost (early stopping at Round 273)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9846	0.5866	0.9836	0.8538	0.8537
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9939	0.9999	0.6172	0.5647	18.0 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.2335	0.2486	0.2408	185	0.9992	0.2506
Backdoor	0.0556	0.0029	0.0054	700	0.9662	0.0270
Benign	1.0000	1.0000	1.0000	333,440	1.0000	1.0000
DoS	0.7583	0.2553	0.3820	897	0.9916	0.4244
Exploits	0.7281	0.7474	0.7376	6,413	0.9957	0.8192
Fuzzers	0.6609	0.8622	0.7483	5,073	0.9964	0.6894
Generic	0.8399	0.8328	0.8364	2,949	0.9982	0.8547
Reconnaissance	0.8076	0.6030	0.6905	2,562	0.9959	0.7389
Shellcode	0.5072	0.7905	0.6179	358	0.9972	0.7175
Worms	0.5484	0.6800	0.6071	25	0.9984	0.6507

E-GraphSAGE (early stopping at Epoch 129)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9819	0.5022	0.9815	0.8279	0.8276
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9539	0.9990	0.4758	0.5097	9.2 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.2570	0.9892	0.4080	185	0.9922	0.2279
Backdoor	0.0067	0.0043	0.0052	700	0.6790	0.0169
Benign	1.0000	0.9998	0.9999	333,440	1.0000	1.0000
DoS	0.3613	0.1918	0.2505	897	0.9843	0.2376

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Exploits	0.7729	0.6379	0.6990	6,413	0.9943	0.7663
Fuzzers	0.6407	0.9074	0.7511	5,073	0.9911	0.6573
Generic	0.7972	0.7664	0.7815	2,949	0.9953	0.8123
Reconnaissance	0.8584	0.5086	0.6387	2,562	0.9935	0.7021
Shellcode	0.2986	0.6173	0.4026	358	0.9180	0.3105
Worms	0.0909	0.0800	0.0851	25	0.9918	0.0275

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
1,918	1,671	71.95%	0.5363	2.9495

HeteroGNN (early stopping at Epoch 67)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9820	0.5074	0.9811	0.8286	0.8284
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9065	0.9977	0.5298	0.5013	9.1 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.4331	0.2973	0.3526	185	0.9891	0.4296
Backdoor	0.0000	0.0000	0.0000	700	0.4007	0.0067
Benign	1.0000	0.9999	1.0000	333,440	1.0000	1.0000
DoS	0.2672	0.1817	0.2163	897	0.9517	0.2074
Exploits	0.7597	0.6304	0.6890	6,413	0.9927	0.7745
Fuzzers	0.6436	0.8533	0.7338	5,073	0.9887	0.6710
Generic	0.7372	0.8494	0.7893	2,949	0.9740	0.8022
Reconnaissance	0.7087	0.5898	0.6438	2,562	0.9914	0.6819
Shellcode	0.2925	0.5866	0.3903	358	0.8157	0.4175
Worms	0.1833	0.4400	0.2588	25	0.9612	0.3076

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
1,703	6,684	100.00%	0.7398	N/A

EGAT (early stopping at Epoch 57)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9697	0.3734	0.9669	0.7153	0.7133
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9915	0.9996	0.4856	0.3997	7.6 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.4606	0.6324	0.5330	185	0.9941	0.6370
Backdoor	0.0000	0.0000	0.0000	700	0.9731	0.0376
Benign	1.0000	0.9994	0.9997	333,440	0.9999	1.0000
DoS	0.0000	0.0000	0.0000	897	0.9868	0.1257
Exploits	0.6515	0.7146	0.6816	6,413	0.9934	0.7408
Fuzzers	0.5810	0.0240	0.0462	5,073	0.9926	0.5453
Generic	0.2437	0.7918	0.3727	2,949	0.9908	0.5102
Reconnaissance	0.9009	0.4863	0.6317	2,562	0.9943	0.6931
Shellcode	0.3341	0.7877	0.4692	358	0.9941	0.5508
Worms	0.0000	0.0000	0.0000	25	0.9956	0.0152

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
1,918	1,671	71.95%	0.5195	N/A

CICIDS — CIC-IDS2018

XGBoost (early stopping at Round 220)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9895	0.6331	0.9863	0.9467	0.9463
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9862	0.9995	0.6443	0.7000	129.2 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Benign	0.9951	0.9996	0.9973	2,608,914	0.9913	0.9987
Bot	0.9998	0.9959	0.9978	16,719	0.9999	0.9970
Brute_Force_-Web	0.0741	0.0082	0.0148	244	0.9941	0.0863
Brute_Force_-XSS	0.0000	0.0000	0.0000	69	0.9998	0.0684
DDoS_attack-HOIC	1.0000	0.9998	0.9999	154,848	1.0000	1.0000
DDoS_attack-LOIC-UDP	0.2449	0.9654	0.3907	260	0.9712	0.2273
DDoS_attacks-LOIC-HTTP	0.9970	0.9818	0.9893	43,289	0.9819	0.9820
DoS_attacks-GoldenEye	0.9729	0.9998	0.9862	4,598	1.0000	0.9999
DoS_attacks-Hulk	0.9979	0.9988	0.9983	15,012	0.9998	0.9996
DoS_attacks-SlowHTTPTest	0.0000	0.0000	0.0000	15,833	0.9950	0.3531
DoS_attacks-Slowloris	0.9813	0.8172	0.8918	2,703	0.9979	0.9317
FTP-BruteForce	0.6469	1.0000	0.7856	29,004	0.9973	0.6469
Infiltration	0.8481	0.3019	0.4453	17,372	0.8644	0.3243
SQL_Injection	0.0000	0.0000	0.0000	66	0.9998	0.0495
SSH-Bruteforce	1.0000	0.9993	0.9996	14,137	1.0000	1.0000

E-GraphSAGE (early stopping at Epoch 151)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9864	0.6380	0.9835	0.9309	0.9305
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9840	0.9994	0.6568	0.6965	79.5 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Benign	0.9931	0.9980	0.9956	2,608,914	0.9842	0.9976
Bot	0.9999	0.9977	0.9988	16,719	0.9996	0.9977
Brute_Force_-Web	0.1554	0.5164	0.2389	244	0.9985	0.1135
Brute_Force_-XSS	0.0000	0.0000	0.0000	69	0.9999	0.2350
DDOS_attack-HOIC	0.9999	1.0000	1.0000	154,848	1.0000	1.0000
DDOS_attack-LOIC-UDP	0.2719	0.9423	0.4220	260	0.9576	0.2234
DDoS_attacks-LOIC-HTTP	1.0000	0.9487	0.9737	43,289	0.9853	0.9847
DoS_attacks-GoldenEye	1.0000	0.7747	0.8730	4,598	0.9997	0.8914
DoS_attacks-Hulk	1.0000	0.8372	0.9114	15,012	1.0000	0.9995
DoS_attacks-SlowHTTPTest	0.0000	0.0000	0.0000	15,833	0.9958	0.4329
DoS_attacks-Slowloris	1.0000	0.8302	0.9072	2,703	0.9999	0.9576
FTP-BruteForce	0.6469	1.0000	0.7856	29,004	0.9977	0.6854
Infiltration	0.5630	0.2811	0.3750	17,372	0.8418	0.2950
SQL_Injection	0.0477	0.6970	0.0893	66	0.9993	0.0376
SSH-Bruteforce	1.0000	1.0000	1.0000	14,137	1.0000	1.0000

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
3,389	3,051	93.69%	0.1686	1.4938

HeteroGNN (early stopping at Epoch 70)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9807	0.6244	0.9804	0.9057	0.9053
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9809	0.9996	0.6847	0.7041	76.8 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Benign	0.9945	0.9903	0.9924	2,608,914	0.9963	0.9992
Bot	0.9999	0.9971	0.9985	16,719	0.9987	0.9977
Brute_Force_-Web	0.7097	0.0902	0.1600	244	0.9898	0.4493
Brute_Force_-XSS	0.0690	0.0290	0.0408	69	0.9740	0.0950

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
DDoS_attack-HOIC	1.0000	1.0000	1.0000	154,848	1.0000	1.0000
DDoS_attack-LOIC-UDP	0.2554	0.9038	0.3983	260	0.9594	0.2034
DDoS_attacks-LOIC-HTTP	1.0000	0.9487	0.9737	43,289	0.9834	0.9806
DoS_attacks-GoldenEye	0.9581	0.7749	0.8568	4,598	1.0000	0.9858
DoS_attacks-Hulk	1.0000	0.8372	0.9114	15,012	1.0000	1.0000
DoS_attacks-SlowHTTPTest	0.0000	0.0000	0.0000	15,833	0.9948	0.3519
DoS_attacks-Slowloris	1.0000	0.8302	0.9072	2,703	1.0000	0.9999
FTP-BruteForce	0.6469	1.0000	0.7856	29,004	0.9972	0.6584
Infiltration	0.2578	0.5066	0.3417	17,372	0.9548	0.4327
SQL_Injection	0.0000	0.0000	0.0000	66	0.8649	0.1161
SSH-Bruteforce	1.0000	1.0000	1.0000	14,137	1.0000	1.0000

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
3,566	12,205	100.00%	0.3714	N/A

EGAT — Training Failed

Training on CICIDS was aborted during the edge-attention update step (GATv2Conv) with a CUDA out-of-memory error — the process had allocated roughly 14.2 GiB against a 14.56 GiB GPU and could not reserve an additional ~700 MiB batch of edge attention weights. No test-set metrics are available for this configuration; UNSW-NB15 training completed without issue on the same hardware.